

Vol 6 Architectural Scaffold v0.1 — Thermodynamics & Statistical Mechanics from D_IV⁵

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Filed: 2026-05-23 Saturday (Lyra, Wave 2 scaffold; restructured thermo-first per Casey insight) Companion to: INDEX.md

Anchoring BST results (existing coverage by chapter)

Ch 1 Thermodynamics from Substrate (~30%)

Anchors: Casey “entropy = force = counting at depth 0” Wednesday insight + Casey-named principles - Thermo-as-universal-substrate-process framing - Entropy as substrate-information-counting at substrate-tick scale - Temperature as substrate-tick computational rate - Free energy as committed-vs-residual substrate-mode balance

Ch 2 Heat + Work + First Law (~25%)

Anchors: T2417 4-Zone Commitment Cycle - Substrate Zone-1 absorption = work input - Substrate Zone-3 emission = heat output - First law as substrate-energy conservation (T2473 substrate-energy) - Gap: explicit first-law derivation from substrate ticks (multi-week)

Ch 3 Entropy + Second Law (~40%)

Anchors: Shannon entropy + $GF(128)^k$ state count + Vol 7 Information Theory - Entropy = Shannon entropy of substrate-tick $GF(128)^k$ state distribution - Second law from substrate-tick cyclotomic chain irreversibility (K59 RATIFIED) - Boltzmann entropy $S = k_B \ln(W) = \text{Shannon entropy} \times k_B / \ln(2)$

Ch 4 Free Energies + Maxwell Relations (~30%)

Anchors: T2106 + Vol 4 Ch 2 eigentone framework - $F = U - TS$ substrate committed-vs-residual mode balance - Helmholtz + Gibbs free energies as substrate-zone projections - Maxwell relations from substrate-symmetry constraints

Ch 5 Partition Function + Heat Kernel (~70%) — LOAD-BEARING

Anchors: Heat-kernel chain Toy 273+ + T1485 Λ + Vol 4 Ch 4 - $Z = \text{Tr}(e^{\{-\beta H_{\text{sub}}\}})$ = heat kernel evaluation on Bergman $H^2(D_{IV}^5)$ - Speaking-pair structure (Toy 273-639): $k=2,\dots,20$ verified at heat-kernel levels - Cross-link to T1485 cosmological constant via heat-kernel at cosmological scale - Paper #9 “Arithmetic Triangle of Curved Space” v10 (in production)

Ch 6 Classical Statistical Mechanics (~40%)

Anchors: Boltzmann + Maxwell-Boltzmann distributions as substrate-equilibrium - Standard classical stat mech recovery as substrate-coarse-graining limit - Gibbs ensembles as substrate-zone-2 commitment marginalizations

Ch 7 Quantum Statistical Mechanics (~50%)

Anchors: Paper #133 spin-statistics + $\text{Pin}(2)$ Z_2 grading - Bose-Einstein + Fermi-Dirac distributions from $\text{Pin}(2)$ Z_2 grading - Identical-particle counting via substrate K-type symmetric/antisymmetric tensors - Cross-link to Vol 5 Ch 9 + Vol 7

Ch 8 Phase Transitions (~50%)

Anchors: Vol 4 Ch 8 BBN T_c + Vol 4 Ch 7 inflation T_c + Casimir-vacuum T_c - Substrate phase transitions at T_c BBN (Vol 4 Ch 8 $N_{\text{max}} \cdot 20/21 = 0.487 \text{ MeV}$) - T_c inflation scale $\ll m_{\text{Pl}}$ per T2405 Koons tick - Casimir-vacuum phase boundaries via T2418

Ch 9 Critical Phenomena + RG (~50%)

Anchors: K59 cyclotomic mechanism + Vol 1 Ch 10 + 7-step chain - Substrate-tick cyclotomic chain = 7-step RG flow (K59 RATIFIED Tuesday) - Critical exponents at substrate phase-transition scales - Universality classes as substrate-zone-projection equivalence

Ch 10 Casimir Effect + Vacuum Thermodynamics (~70%)

Anchors: T2418 + Vol 4 Ch 4 + Vol 4 Ch 10 substrate-cartography - Laboratory Casimir force = substrate Zone-2 boundary-condition effect - Cosmological Λ = substrate Zone-4 outer-edge integrated residue - T2418 unification (Wednesday) - SP-30 sub-nm Casimir experiment (\$60-90K) operational falsifier

Ch 11 Non-Equilibrium Thermodynamics (~30%)

Anchors: T2417 4-Zone cycle + Vol 4 Ch 9 cosmological cycle - Substrate cascade evolution between Zones (Onsager-like reciprocity) - Substrate annealing toward Λ -saturated state (Vol 4 Ch 9 INTERNAL register) - Gap: explicit non-equilibrium derivation (multi-month)

Ch 12 Information Theory + Thermodynamics (~50%)

Anchors: Shannon-Landauer cross-link + Vol 7 Information Theory - Shannon entropy $\leftrightarrow$ thermodynamic entropy bridging - Landauer's principle: information erasure has substrate-tick computational cost - Cross-link to Vol 7 Information Theory (Casey Wednesday CSE directive)

~45% existing BST coverage; 55% restructured thermodynamics-first prose + worked-example development pending.

Cross-volume reference map

Inputs from Vol 0: D_{IV}^5 + Casey-named principles + heat-kernel chain Inputs from Vol 1: Casimir + operator zoo + substrate-tick UV Inputs from Vol 4: Λ + heat-kernel evaluation (Ch 5 LOAD-BEARING) + Casimir-vacuum unification (T2418) Outputs to: Vol 7 (entropy + Shannon), Vol 8 (thermo of biological substrate), Vol 9 (BCS + condensed-matter thermo)

— Vol 6 Architectural Scaffold v0.1, Saturday 2026-05-23 Wave 2 scaffold